

Lab 2: Sampling, FIR Filtering, and Time-Frequency Analysis

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1. Sampling parameters

For phase $\phi(t)$ in radians, instantaneous frequency is $\phi'(t)/(2\pi)$. The lab comparison uses $F_s = 2f_{\max}$, $4f_{\max}$, and $20f_{\max}$ for a 1 s record. These rates are based on maximum instantaneous frequency. For modulated and finite-duration signals, this need not bound the full spectrum.

```
lab2_signalParameters;
```

Signal	MaxInstFreq_Hz	Fs_Nyquist_Hz	Fs_2Nyquist_Hz	Fs_10Nyquist_Hz
{'Sinusoid' }	10	20	40	200
{'Linear chirp' }	30	60	120	600
{'Quadratic chirp' }	25	50	100	500
{'Transient linear chirp' }	30	60	120	600
{'Sine-Gaussian' }	30	60	120	600
{'AM' }	40	80	160	800
{'FM' }	60	120	240	1200
{'AM-FM' }	60	120	240	1200

1.1 Sinusoid

The signal has a constant frequency of 10 Hz. At $F_s=20$ Hz, a zero-phase sine is sampled at its zero crossings and disappears. Sampling exactly at the Nyquist boundary can therefore lose phase-dependent information.

```
reportNyquist(1);
```

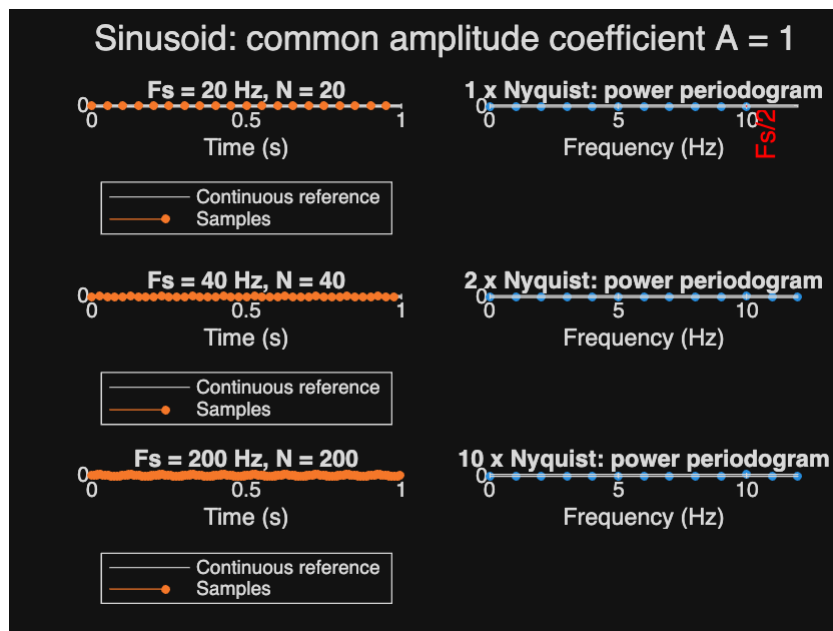
Signal	MaxInstFreq_Hz	Fs_Nyquist_Hz	Fs_2Nyquist_Hz	Fs_10Nyquist_Hz
{'Sinusoid' }	10	20	40	200
{'Linear chirp' }	30	60	120	600
{'Quadratic chirp' }	25	50	100	500
{'Transient linear chirp' }	30	60	120	600
{'Sine-Gaussian' }	30	60	120	600
{'AM' }	40	80	160	800
{'FM' }	60	120	240	1200
{'AM-FM' }	60	120	240	1200

$F_s = 20$ Hz: zero-phase carrier samples are zero at Nyquist.

$F_s = 20$ Hz, $N = 20$, mean square = 0.000000, PSD integral = 0.000000

$F_s = 40$ Hz, $N = 40$, mean square = 0.500000, PSD integral = 0.500000

$F_s = 200$ Hz, $N = 200$, mean square = 0.500000, PSD integral = 0.500000



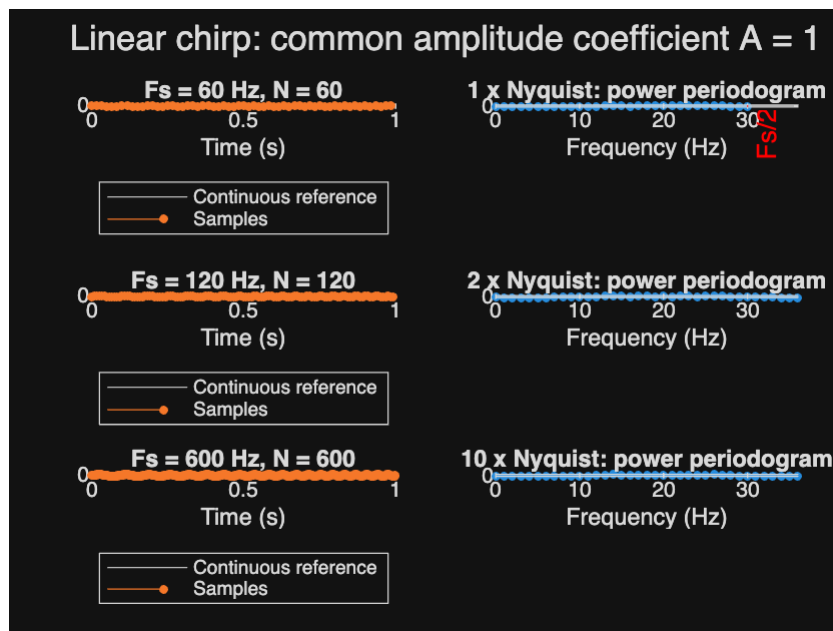
1.2 Linear chirp

The instantaneous frequency rises from 10 to 30 Hz. Higher sampling rates describe the waveform more faithfully. The spectrum spans a frequency interval because the signal sweeps through multiple frequencies.

```
reportNyquist(2);
```

Signal	MaxInstFreq_Hz	Fs_Nyquist_Hz	Fs_2Nyquist_Hz	Fs_10Nyquist_Hz
{'Sinusoid' }	10	20	40	200
{'Linear chirp' }	30	60	120	600
{'Quadratic chirp' }	25	50	100	500
{'Transient linear chirp' }	30	60	120	600
{'Sine-Gaussian' }	30	60	120	600
{'AM' }	40	80	160	800
{'FM' }	60	120	240	1200
{'AM-FM' }	60	120	240	1200

$F_s = 60$ Hz, $N = 60$, mean square = 0.471993, PSD integral = 0.471993
 $F_s = 120$ Hz, $N = 120$, mean square = 0.499937, PSD integral = 0.499937
 $F_s = 600$ Hz, $N = 600$, mean square = 0.499939, PSD integral = 0.499939



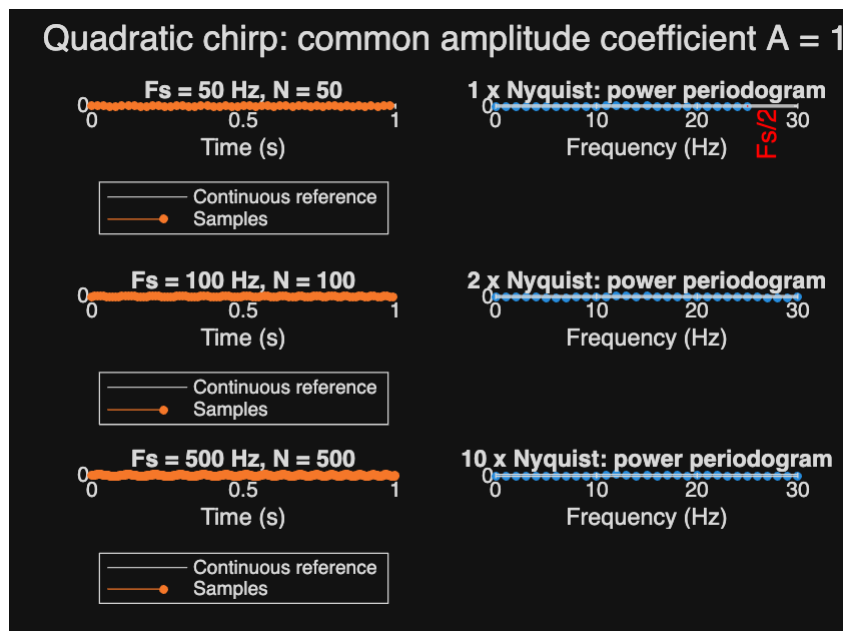
1.3 Quadratic chirp

The frequency is $10+6*t+9*t^2$ Hz and reaches 25 Hz at $t=1$ s. The sweep accelerates, so the signal spends unequal amounts of time at each frequency.

```
reportNyquist(3);
```

Signal	MaxInstFreq_Hz	Fs_Nyquist_Hz	Fs_2Nyquist_Hz	Fs_10Nyquist_Hz
{'Sinusoid' }	10	20	40	200
{'Linear chirp' }	30	60	120	600
{'Quadratic chirp' }	25	50	100	500
{'Transient linear chirp' }	30	60	120	600
{'Sine-Gaussian' }	30	60	120	600
{'AM' }	40	80	160	800
{'FM' }	60	120	240	1200
{'AM-FM' }	60	120	240	1200

$F_s = 50$ Hz, $N = 50$, mean square = 0.474482, PSD integral = 0.474482
 $F_s = 100$ Hz, $N = 100$, mean square = 0.499981, PSD integral = 0.499981
 $F_s = 500$ Hz, $N = 500$, mean square = 0.499986, PSD integral = 0.499986



1.4 Transient linear chirp

During $0.2 \leq t \leq 0.7$ s, frequency rises from 10 to 30 Hz. The finite gate reduces record-averaged power and introduces spectral spreading.

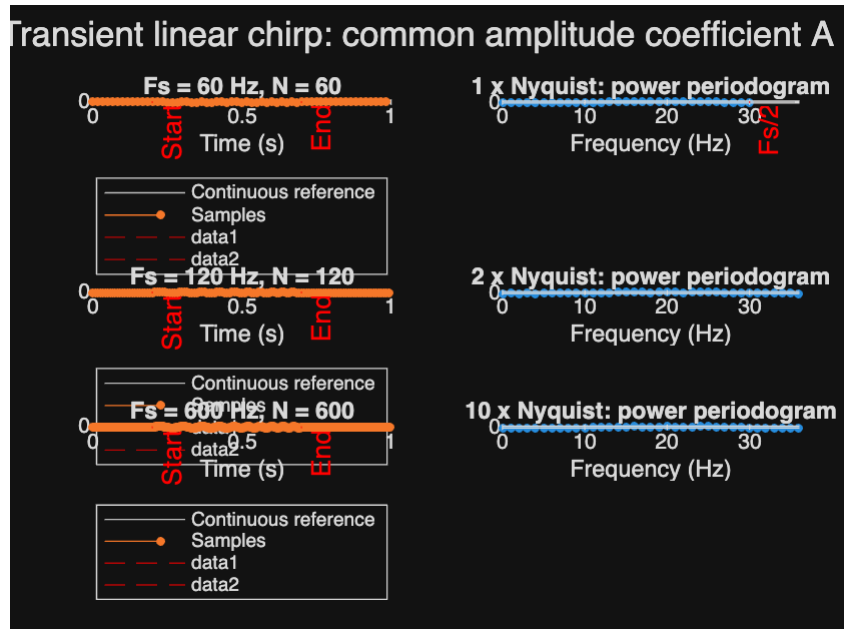
```
reportNyquist(4);
```

Signal	MaxInstFreq_Hz	Fs_Nyquist_Hz	Fs_2Nyquist_Hz	Fs_10Nyquist_Hz
{'Sinusoid' }	10	20	40	200
{'Linear chirp' }	30	60	120	600
{'Quadratic chirp' }	25	50	100	500
{'Transient linear chirp' }	30	60	120	600
{'Sine-Gaussian' }	30	60	120	600
{'AM' }	40	80	160	800
{'FM' }	60	120	240	1200
{'AM-FM' }	60	120	240	1200

$F_s = 60$ Hz, $N = 60$, mean square = 0.230126, PSD integral = 0.230126

$F_s = 120$ Hz, $N = 120$, mean square = 0.249876, PSD integral = 0.249876

$F_s = 600$ Hz, $N = 600$, mean square = 0.249880, PSD integral = 0.249880



1.5 Sine-Gaussian

A Gaussian envelope localizes a 30 Hz carrier around $t=0.5$ s. At $F_s=60$ Hz, the zero-phase carrier samples vanish. The envelope broadens the spectrum, so twice the carrier frequency is not a strict bandwidth-based Nyquist rate.

```
reportNyquist(5);
```

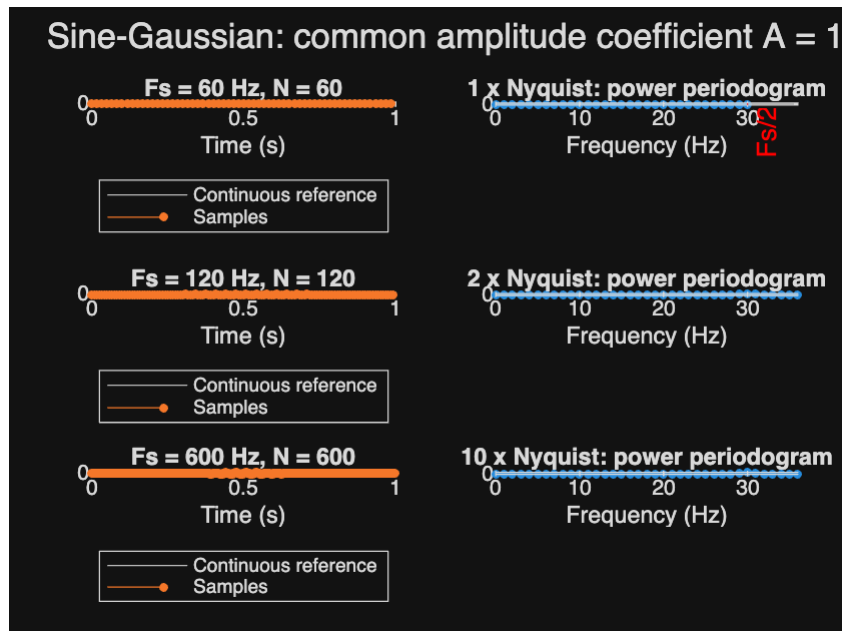
Signal	MaxInstFreq_Hz	Fs_Nyquist_Hz	Fs_2Nyquist_Hz	Fs_10Nyquist_Hz
{'Sinusoid' }	10	20	40	200
{'Linear chirp' }	30	60	120	600
{'Quadratic chirp' }	25	50	100	500
{'Transient linear chirp' }	30	60	120	600
{'Sine-Gaussian' }	30	60	120	600
{'AM' }	40	80	160	800
{'FM' }	60	120	240	1200
{'AM-FM' }	60	120	240	1200

$F_s = 60$ Hz: zero-phase carrier samples are zero at Nyquist.

$F_s = 60$ Hz, $N = 60$, mean square = 0.000000, PSD integral = 0.000000

$F_s = 120$ Hz, $N = 120$, mean square = 0.088623, PSD integral = 0.088623

$F_s = 600$ Hz, $N = 600$, mean square = 0.088623, PSD integral = 0.088623



1.6 Amplitude modulation

$\cos(2\pi \cdot 5 \cdot t) \cdot \sin(2\pi \cdot 40 \cdot t)$ produces equal-amplitude sidebands at 35 and 45 Hz. The carrier-based rate of 80 Hz misses the upper sideband and gives zero samples for this phase. The actual two-tone Nyquist boundary is 90 Hz.

```
reportNyquist(6);
```

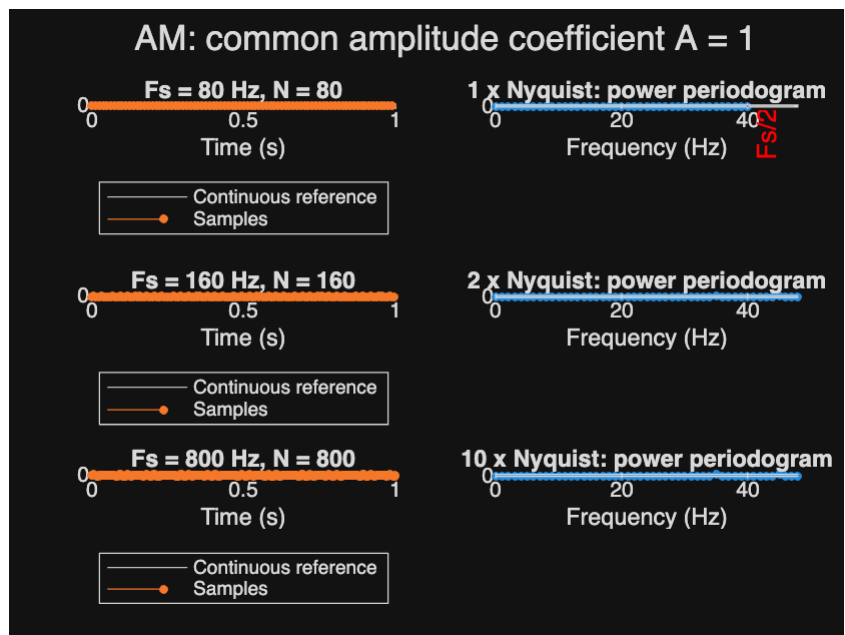
Signal	MaxInstFreq_Hz	Fs_Nyquist_Hz	Fs_2Nyquist_Hz	Fs_10Nyquist_Hz
{'Sinusoid' }	10	20	40	200
{'Linear chirp' }	30	60	120	600
{'Quadratic chirp' }	25	50	100	500
{'Transient linear chirp' }	30	60	120	600
{'Sine-Gaussian' }	30	60	120	600
{'AM' }	40	80	160	800
{'FM' }	60	120	240	1200
{'AM-FM' }	60	120	240	1200

Fs = 80 Hz: zero-phase carrier samples are zero at Nyquist.

Fs = 80 Hz, N = 80, mean square = 0.000000, PSD integral = 0.000000

Fs = 160 Hz, N = 160, mean square = 0.250000, PSD integral = 0.250000

Fs = 800 Hz, N = 800, mean square = 0.250000, PSD integral = 0.250000



1.7 Frequency modulation

The instantaneous frequency is $40 - 20 \sin(2\pi \cdot 5 \cdot t)$ Hz, ranging from 20 to 60 Hz. FM has sidebands beyond that range; twice the maximum instantaneous frequency is not an exact alias-free sampling guarantee.

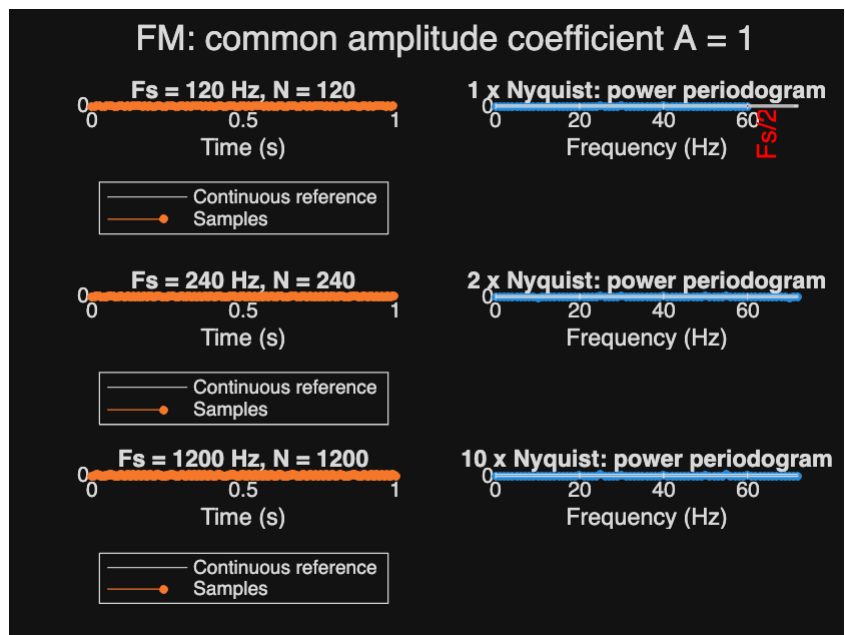
```
reportNyquist(7);
```

Signal	MaxInstFreq_Hz	Fs_Nyquist_Hz	Fs_2Nyquist_Hz	Fs_10Nyquist_Hz
{'Sinusoid' }	10	20	40	200
{'Linear chirp' }	30	60	120	600
{'Quadratic chirp' }	25	50	100	500
{'Transient linear chirp' }	30	60	120	600
{'Sine-Gaussian' }	30	60	120	600
{'AM' }	40	80	160	800
{'FM' }	60	120	240	1200
{'AM-FM' }	60	120	240	1200

$F_s = 120$ Hz, $N = 120$, mean square = 0.388234, PSD integral = 0.388234

$F_s = 240$ Hz, $N = 240$, mean square = 0.499961, PSD integral = 0.499961

$F_s = 1200$ Hz, $N = 1200$, mean square = 0.499961, PSD integral = 0.499961



1.8 Combined AM-FM

The envelope and instantaneous frequency both change periodically. The amplitude modulation shifts the FM sidebands by ± 5 Hz. Higher sampling rates improve the representation of the significant spectral components.

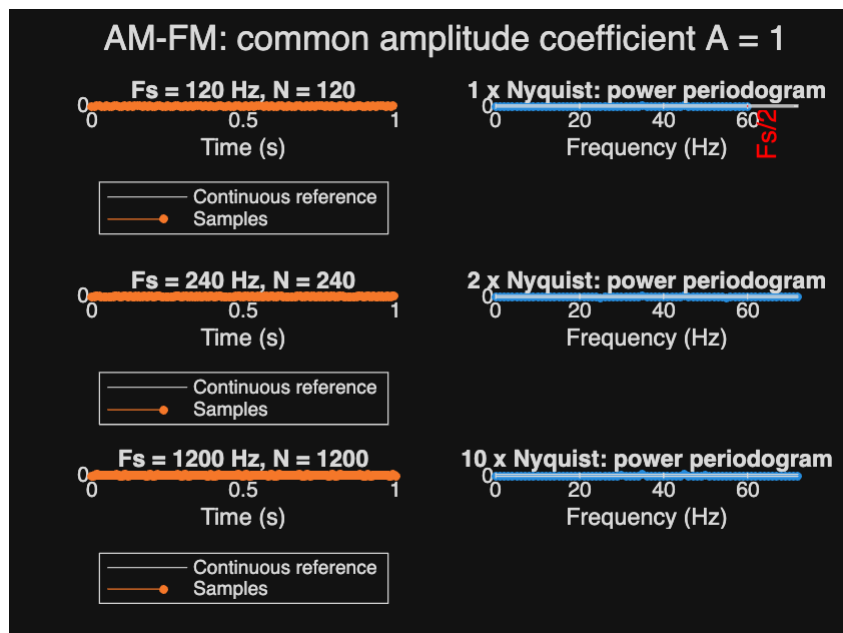
```
reportNyquist(8);
```

Signal	MaxInstFreq_Hz	Fs_Nyquist_Hz	Fs_2Nyquist_Hz	Fs_10Nyquist_Hz
{'Sinusoid' }	10	20	40	200
{'Linear chirp' }	30	60	120	600
{'Quadratic chirp' }	25	50	100	500
{'Transient linear chirp' }	30	60	120	600
{'Sine-Gaussian' }	30	60	120	600
{'AM' }	40	80	160	800
{'FM' }	60	120	240	1200
{'AM-FM' }	60	120	240	1200

Fs = 120 Hz, N = 120, mean square = 0.244038, PSD integral = 0.244038

Fs = 240 Hz, N = 240, mean square = 0.250108, PSD integral = 0.250108

Fs = 1200 Hz, N = 1200, mean square = 0.250108, PSD integral = 0.250108



Sampling and PSD interpretation

All records last 1 s, so their DFT spacing remains approximately 1 Hz. Increasing F_s increases sample density and Nyquist frequency, not frequency resolution at fixed duration. The one-sided PSD uses $\text{FFT}^2/(F_s \cdot N)$, with doubled interior positive-frequency bins. Summing $\text{PSD} \cdot df$ equals $\text{mean}(x.^2)$. This Parseval check validates normalization, not absence of aliasing.

2. Three-component signal and FIR filters

The input contains 100, 200, and 300 Hz sinusoids with amplitudes 10, 5, and 2.5, and phases 0, $\pi/6$, and $\pi/4$. With $N=2048$ and $F_s=1024 \text{ Hz}$, duration is 2 s, $df=0.5 \text{ Hz}$, and Nyquist frequency is 512 Hz. Component mean-square powers are 50, 12.5, and 3.125: total 65.625. Peak PSD values are 100, 25, and 6.25 because each occupies a 0.5 Hz bin.

Order-100 Hamming-window FIR filters use cutoffs of 150 Hz (lowpass), 150-250 Hz (bandpass), and 250 Hz (highpass). These lie between the tones. The 101 symmetric coefficients give a delay of 50 samples, or 0.048828 s.

```
lab2_filtering;
```

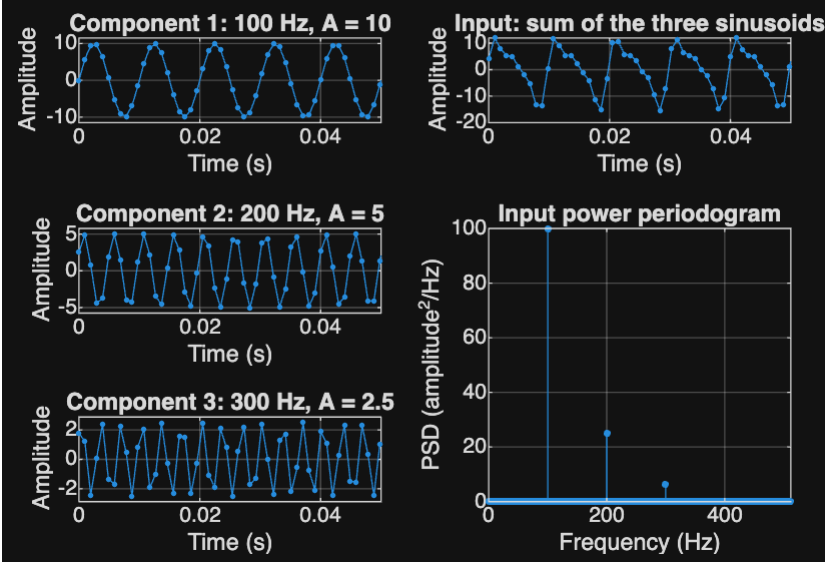
Frequency_Hz	Amplitude	Phase_rad	MeanSquare	PeakPSD
100	10	0	50	100
200	5	0.5236	12.5	25
300	2.5	0.7854	3.125	6.25

$N = 2048$, $F_s = 1024 \text{ Hz}$, $T = 2 \text{ s}$, $df = 0.5 \text{ Hz}$

Nyquist frequency = 512 Hz

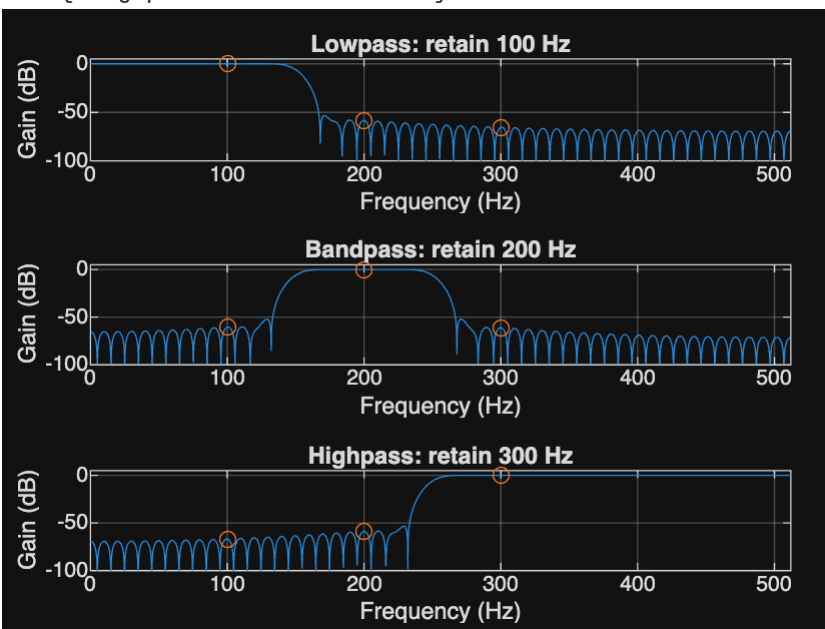
Input mean square = 65.625000; integrated PSD = 65.625000

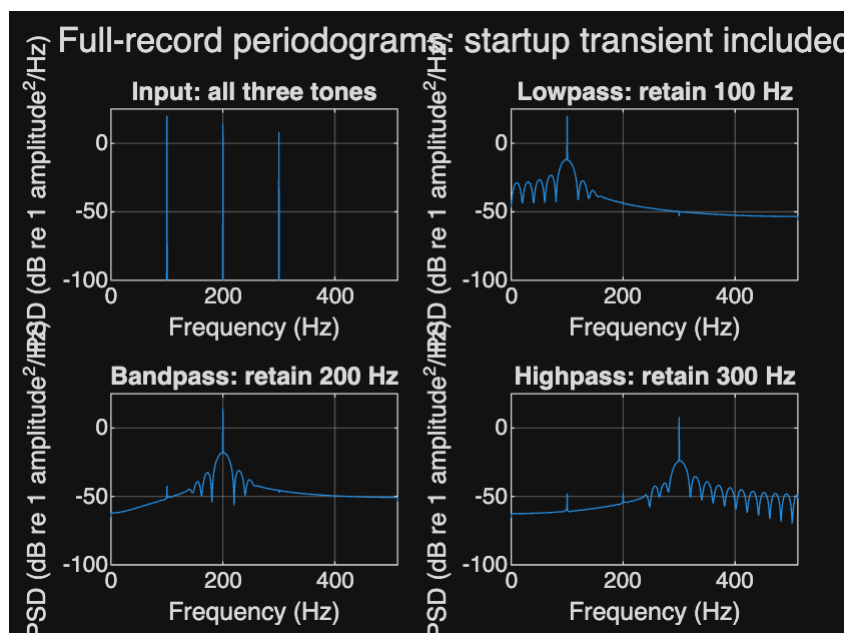
Filtering stage 1: verify the input before designing filter



FIR order = 100, taps = 101, group delay = 50 samples (0.048828 s)

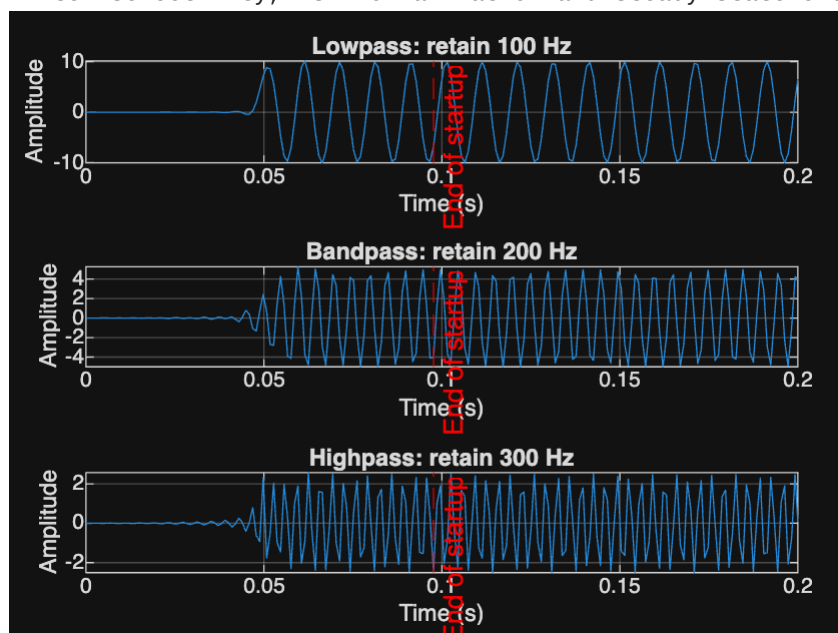
Filter	Gain100Hz_dB	Gain200Hz_dB	Gain300Hz_dB
{'Lowpass: retain 100 Hz' }	0.0045397	-58.312	-65.489
{'Bandpass: retain 200 Hz'}	-60.507	5.786e-15	-61.104
{'Highpass: retain 300 Hz'}	-66.747	-58.833	0.017589





Filter	Amplitude100Hz	Amplitude200Hz	Amplitude300Hz	PredictedSteadyPo
{'Lowpass: retain 100 Hz' }	10.005	0.0060725	0.0013289	50.052
{'Bandpass: retain 200 Hz'}	0.0094327	5	0.0022016	12.5
{'Highpass: retain 300 Hz'}	0.0045989	0.0057192	2.5051	3.1377

Filter selectivity, PSD normalization and steady-state checks passed.



Filtering results and physical interpretation

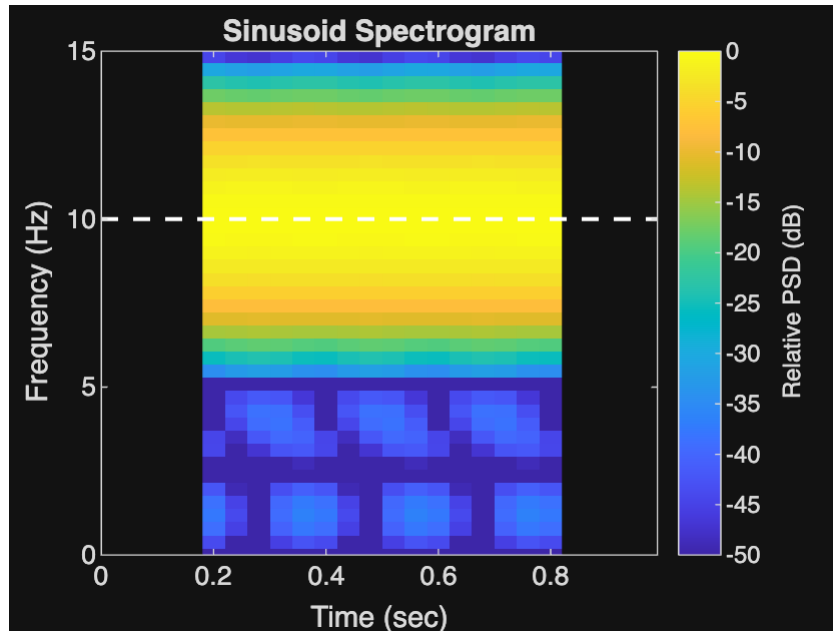
Lowpass retains the 100 Hz component with amplitude about 10.005. Bandpass retains the 200 Hz component with amplitude 5. Highpass retains the 300 Hz component with amplitude about 2.505. Unwanted tones are attenuated by roughly 58-67 dB. Small passband gain deviations are expected for these finite-order filters. Full-record output PSDs include startup transients, producing broad leakage. After the first 100 samples,

predicted and computed waveforms agree to approximately $1e-12$. Numerical checks verify selectivity and PSD scaling.

3.1 Sinusoid spectrogram

A stationary horizontal ridge is centered at 10 Hz. Its finite width comes from the analysis window; the original signal is a single tone.

```
reportSpectrogram('testcrbogensig');
```

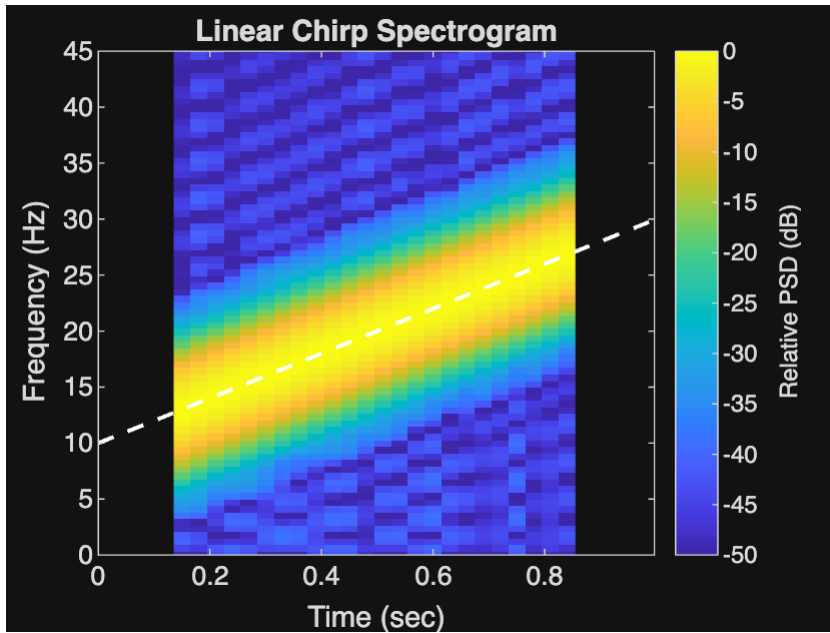


Spectrogram: window = 40 samples (0.400 s), overlap = 36, nfft = 256
Time step = 0.040 s; frequency grid spacing = 0.391 Hz

3.2 Linear chirp spectrogram

The rising ridge follows $f(t)=10+20*t$ Hz and directly displays the sweep.

```
reportSpectrogram('testcrbgenlcsig');
```

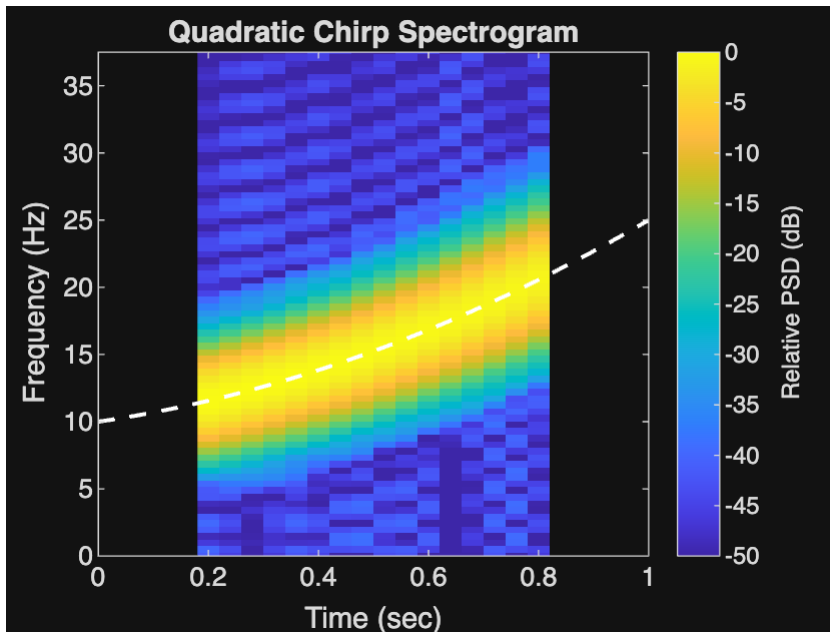


Spectrogram: window = 90 samples (0.300 s), overlap = 81, nfft = 512
Time step = 0.030 s; frequency grid spacing = 0.586 Hz

3.3 Quadratic chirp spectrogram

The curved ridge follows $f(t)=10+6t+9t^2$ Hz, showing an accelerating sweep.

```
reportSpectrogram('testcrbgenqcsig');
```

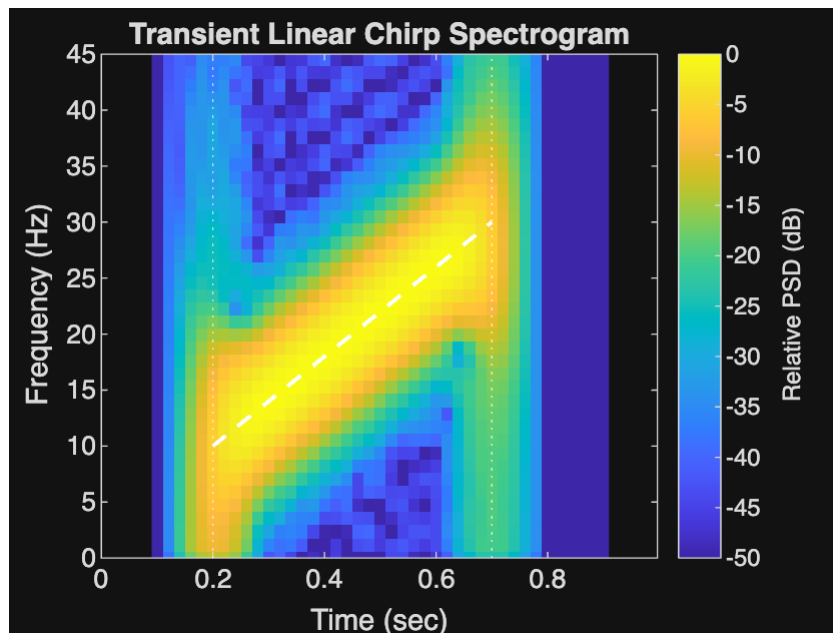


Spectrogram: window = 100 samples (0.400 s), overlap = 90, nfft = 512
Time step = 0.040 s; frequency grid spacing = 0.488 Hz

3.4 Transient chirp spectrogram

Power is concentrated during the gate interval. Windows overlapping its boundaries smear timing, and abrupt gating introduces broadband content.

```
reportSpectrogram('testcrbgenltcsig');
```

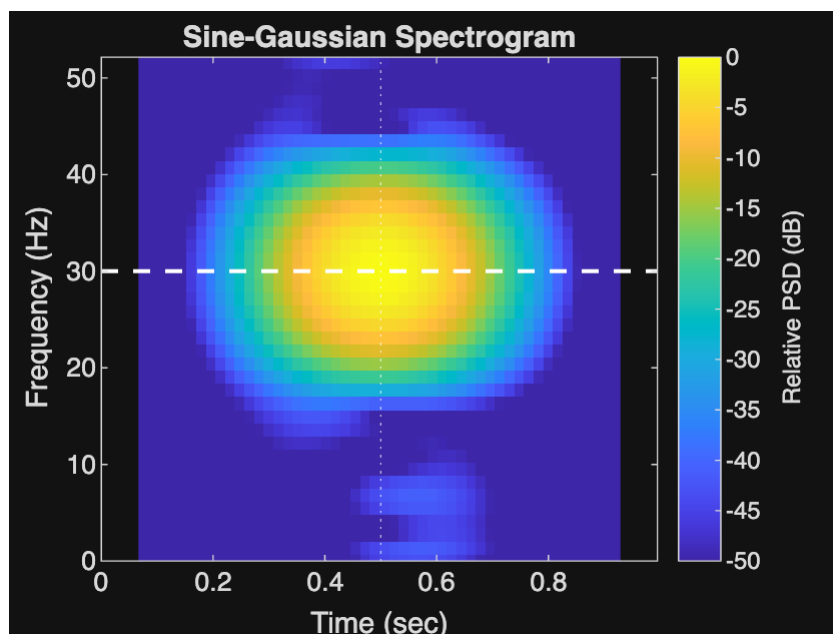


Spectrogram: window = 60 samples (0.200 s), overlap = 54, nfft = 256
Time step = 0.020 s; frequency grid spacing = 1.172 Hz

3.5 Sine-Gaussian spectrogram

The time-frequency patch is centered near 0.5 s and 30 Hz. Localization comes from the Gaussian pulse, with extra broadening from the STFT window.

```
reportSpectrogram('testcrbgensgsig');
```

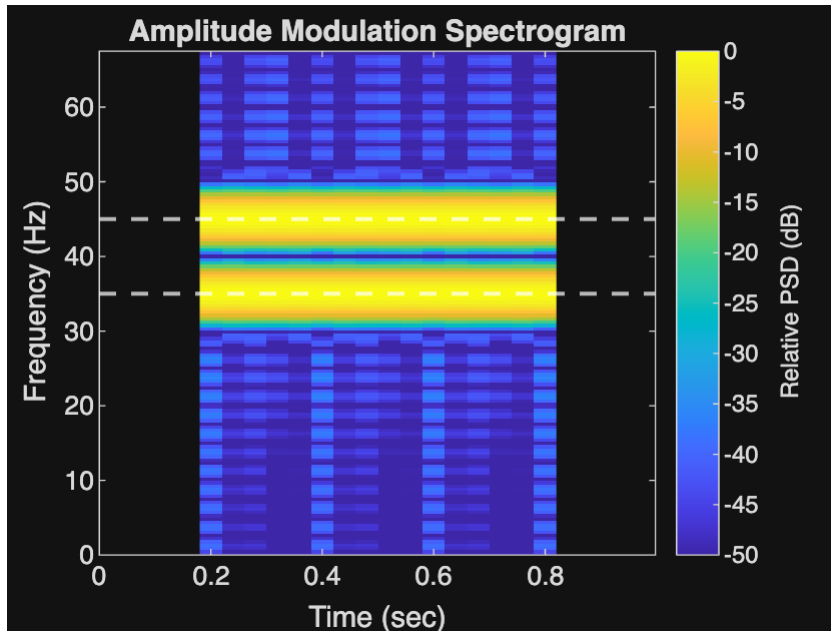


Spectrogram: window = 52 samples (0.150 s), overlap = 46, nfft = 256
Time step = 0.017 s; frequency grid spacing = 1.358 Hz

3.6 AM spectrogram

Horizontal ridges at 35 and 45 Hz verify the suppressed-carrier sidebands. The 0.4 s window emphasizes frequency separation and averages the envelope.

```
reportSpectrogram('testcrcbgenamsig');
```

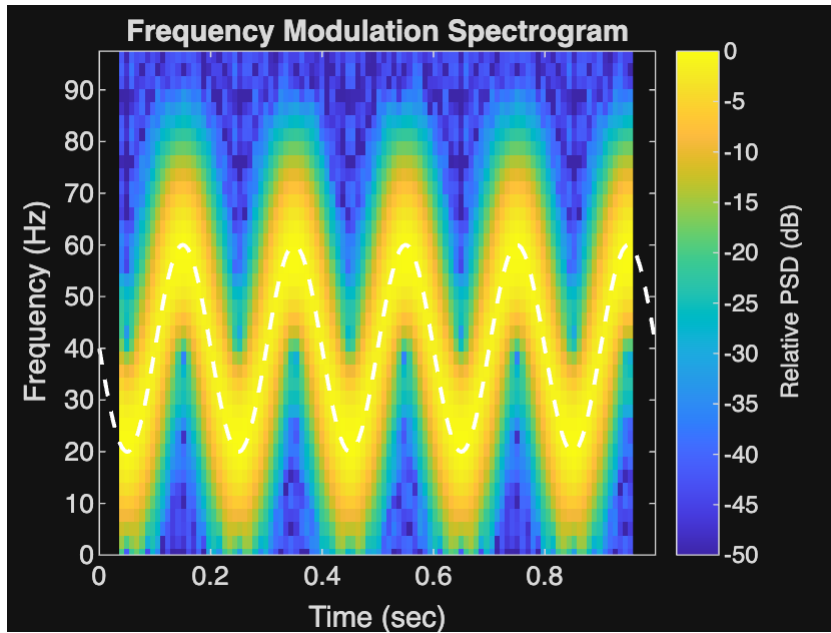


Spectrogram: window = 180 samples (0.400 s), overlap = 162, nfft = 1024
Time step = 0.040 s; frequency grid spacing = 0.439 Hz

3.7 FM spectrogram

A periodic ridge follows the 20-60 Hz instantaneous-frequency variation. The 0.08 s window reveals rapid changes but produces a broad frequency band.

```
reportSpectrogram('testcrcbgenfmsig');
```

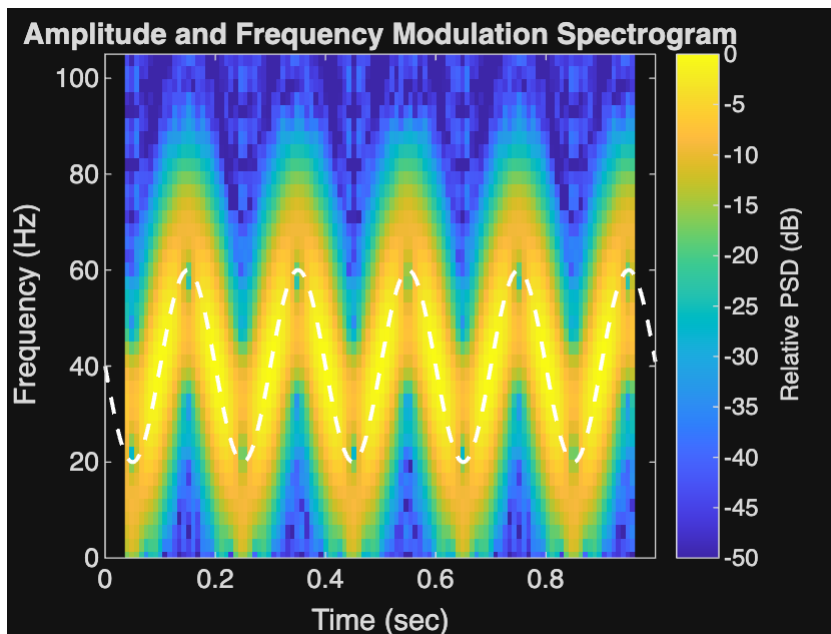


Spectrogram: window = 52 samples (0.080 s), overlap = 46, nfft = 256
Time step = 0.009 s; frequency grid spacing = 2.539 Hz

3.8 AM-FM spectrogram

Frequency varies periodically while the envelope changes power. Short windows reveal time variation, although frequency detail is less resolved.

```
reportSpectrogram('testcrgbgenamfmsig');
```



Spectrogram: window = 56 samples (0.080 s), overlap = 50, nfft = 256
Time step = 0.009 s; frequency grid spacing = 2.734 Hz

Conclusions

Sampling at the instantaneous-frequency-based boundary can distort or even erase a signal. Bandwidth and phase must also be considered. FIR filters isolate the three tones with predictable delay and strong rejection. Spectrograms distinguish stationary tones, sweeps, localized pulses, and modulation. Window duration controls the time-frequency resolution tradeoff; overlap increases time-grid density, while zero padding refines the frequency grid without improving physical frequency resolution. Color is relative to each plot's maximum and cannot be used to compare absolute power across plots.

Local report helpers

```
function reportNyquist(selectedSignal)
    sourcePath = which('lab2_nyquistComparison');
    assert(~isempty(sourcePath), 'Place lab2_nyquistComparison.m on the
path. ');
    sourceText = fileread(sourcePath);
    selector = 'signalIdx\s*=\s*\d+\s*';
    assert(numel(regexps(sourceText, selector))==1, ...
        'Expected one signal selector in lab2_nyquistComparison.m. ');
    sourceText = regexprep(sourceText, selector, ...
        sprintf('signalIdx = %d;', selectedSignal));
    eval(sourceText);
end

function reportSpectrogram(testName)
    sourcePath = which(testName);
    assert(~isempty(sourcePath), 'Missing test script: %s', testName);
    sourceText = fileread(sourcePath);
    firstPlot = strfind(sourceText, '% Plot the sampled signal');
    specSection = strfind(sourceText, '% Spectrogram');
    assert(~isempty(firstPlot) && ~isempty(specSection), ...
        'Use the updated Lab 2 test script for %s.', testName);
    % Generate the original signal, then execute only its spectrogram block.
    % Local function workspace prevents variables from leaking between tests.
    eval(sourceText(1:firstPlot(1)-1));
    eval(sourceText(specSection(1):end));
end
```